

The Kinected DJ

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ABSTRACT

This paper introduces the Kinected DJ: a prototype for a digital music instrument that tracks full body motion and uses the tracked information to modify audio effect parameters. Related work is discussed, followed by an analysis of the required software, hardware, and programming for the Kinected DJ. Implementation and future work are also addressed to best evaluate the effectiveness of the design and where it may have room for improvement.

1. INTRODUCTION

Digital musical instruments, originally conceptualized in the late 20th century, are extremely diverse in the way they create new instruments and adapt pre-existing ones, expanding sonic horizons with each new iteration. With the advent of motion tracking technologies like the Xbox Kinect and LeapMotion, tracking the human body is cheaper and more accessible than ever before, allowing for DMIs that focus specifically on integrating the body into performance in ways that previously seemed impossible. DSP software like Max/MSP and PureData have also facilitated audio programming, allowing many budding instrument designers to try their hand at making novel systems and sounds.

The Kinected DJ is an example of this; a system that tracks full-body movements of a performer to control audio effects during a DJ set, this instrument was realized using the Xbox Kinect, KinectV2-OSC (a program that decodes tracking data), and Max/MSP to create and manipulate audio effects. Also necessary in this device is a DJ controller, though the controller is up to the user's discretion. The goal of this system is to expand the scope of motion tracking DMIs to include different types of tracking and performance.

2. RELATED WORK

Motion tracking has been used in the creation of digital musical instruments since the early 2010s, with various DMIs using tracking strategies as diverse as the Xbox Kinect and LeapMotion [1][4][6][7], infrared blob detection [3], FaceOSC [9], and OptiTrack [10]. Many past instruments have focused on tracking specifically the arms and hands [1][3][4][5][6][7][10], while others have tracked other body parts [9], or even inanimate objects [1][2][8].

The reason for hand-centric tracking lies primarily in that most of the past instruments were either extending the piano or creating a new piano-like instrument [1][2][3][6][10]. A wide variety of software was also used in the creation of these instruments, including Max/MSP [1][7][8], PureData [3][5], and AAAmbi [1][2], among others. Many of these past projects have included performances [4][6][9] or user studies [5][10] and have as a result come to conclusions about possible improvements to their design.

These projects have been invaluable to the development of new modes of artistic expression, and the hope of the Kinected DJ project is to build on the work that these authors have begun. Having full-body tracking is a unique design choice, the goal of this being that users have an even greater degree of artistic freedom to experiment with this relatively new technology. Additionally, there have been few, if any, past DMIs that extend DJ performances, and it is hoped that the Kinected DJ will fill this hole in the world of new instrument design.

3. DESIGN AND SETUP

A main goal for this device is the ability to make use of the emotive capacity of the full human body to extend the possibilities of a DJ performance. This was desirable as it would allow the performer to be more mobile and interesting to watch during their performance. Rather than having to stand still and turn small knobs to change sound effect parameters, the Kinected DJ would allow the performer to move different parts of their body in 3D space, creating a more stimulating and compelling visual performance, with the added benefit of having more varied audio effects than most DJ controllers.

3.1 DJ Controller and Software

While the Kinected DJ can be sent any audio source to manipulate, it is primarily intended to work in conjunction with pre-existing DJ hardware and software to allow for a hybrid DJ performance. While the hardware and software are interchangeable, the Rekordbox software and the DDJ-400 DJ controller were used for this iteration of the Kinected DJ. The software allows users to upload audio files or browse songs from streaming services like SoundCloud and Tidal, after which they can play multiple files on different tracks in the software while also applying effects. The DDJ controller allows these effects to be manipulated with a physical controller rather than a UI,

simplifying the process of mixing tracks during a performance.



Figure 2. Rekordbox UI

While the Kinected DJ provides a much greater variety of effects and a more dynamic system for implementing them, Rekordbox makes it easy to generate diverse mixes on the fly, as users can pick from whatever song they like. This makes the DDJ controller desirable as it minimizes the need to click through computer menus, instead allowing the performer to move through songs quickly and easily without being unduly distracted from the physical performance that the Kinected DJ enables.

3.2 Motion Tracking

Motion tracking was implemented with an Xbox Kinect in conjunction with KinectV2-OSC, a software that decodes tracking data for multiple body joints, giving X, Y, and Z positions as outputs. The software can also track the opening and closing of each hand, representing this with and open or closed output state. These outputs can then be sent to a desired IP address via a specific OSC port number that can be changed by a text file in the KinectV2-OSC folder. KinectV2-OSC is open source, and a full download is accessible through GitHub.

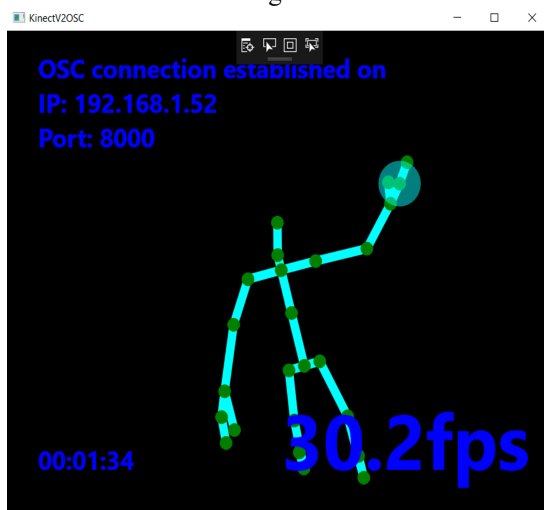


Figure 1. KinectV2-OSC UI

3.3 Max/MSP Programming

Once the position data is transmitted via OSC, it is received by Max/MSP, where it is then mapped to audio effect parameters that can be implemented on an audio signal. To do this, the OSC messages must first be routed to specific outputs for each joint of the body, after which each piece of X, Y, and Z position data can be calibrated. Calibration is achieved by setting a low and high position value by moving each body part along each positional axis and then scaling the inputs for more consistent output data when targeting audio effect parameters. Once this is achieved, a wide variety of effect parameters can be individually connected to an axis of positional data for any given joint. For discrete effect parameters, opening and closing the hands can be linked to cycle through desirable effect options.

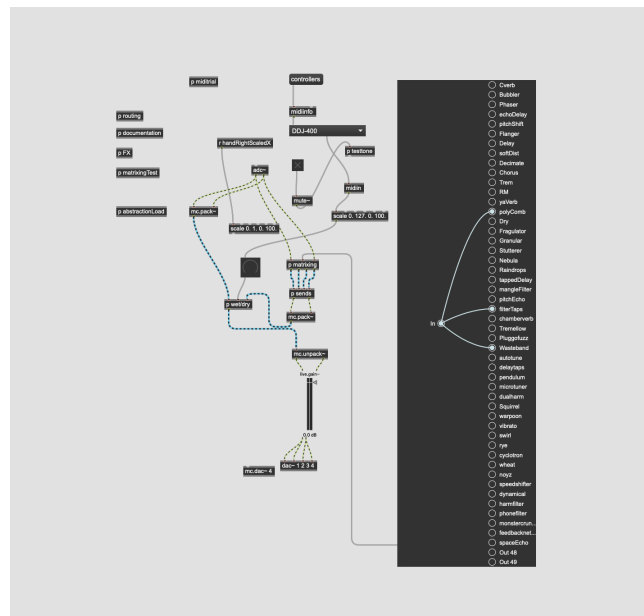


Figure 3. Kinected DJ Max/MSP interface with effect patcher

This iteration of the Kinected DJ has 48 effects (as shown in Figure 3), each of which can be toggled on or off through a Max patcher. This allows a huge variety of effect combinations as different effects can be mixed and matched at the user's discretion. There is also a master wet/dry knob which can be connected to various MIDI inputs to make this control more user-friendly.

3.4 THE KINECTED DJ

Beyond the previously mentioned hardware and software, two laptops were used in this iteration of the Kinected DJ. One laptop was a Windows machine that could run KinectV2-OSC, and the other was a MacBook that could run Rekordbox and Max/MSP. When connected on the same network, the Windows computer sends outputs via

OSC to the Mac which are received in Max. Rekordbox is then configured to send audio into Max, where audio undergoes manipulation based on motion tracking inputs, after which it sends the audio back out to the DDJ controller to be output to speakers.



Figure 4. The full Kinected DJ setup including MIDI controller, laptops for running Max, Rekordbox, and Kinect V2 OSC, DJ controller, and Kinect.

One benefit of the Kinected DJ is its flexible implementation, allowing for different computer setups (all the programs could be run from one Windows computer) as well as a diverse set of DJ software and DJ controllers. Even pre-recorded DJ sets, or perhaps longer individual pieces, could be sent to Max for processing during the performance, allowing for use of this device beyond the intended hybrid DJ performance.

Microphone inputs from acoustic instrument performances could also be modified by the Kinected DJ and allow for extended options for musical expression. This would be ideal for standing performances, such as singing or playing wind and string instruments, as the Kinect operates best when tracking a standing subject.

4. Evaluation

Aside from its flexibility, there are several benefits- and places for improvement- to the Kinected DJ. First, its diverse selection of inputs allows for a great deal of customizability when choosing mappings, meaning with basic programming knowledge a user can send the motion of any specific body part to nearly any effect parameter they desire. With practice, this allows a performer to have full control over which effect is used with which motion.

Additionally, the performer can easily add their own effects and link them to the routing with only minor scaling necessary. Together, these benefits allow a user to modify the instrument to their own specifications, allowing for a wide array of new sounds and performance modes to

be explored. The Kinected DJ is also fairly affordable and easy to transport, requiring only a laptop, a DJ controller, a Kinect, and Max/MSP (as well as a few free applications).

Despite this, one benefit to the Kinected DJ is also a drawback: performers may feel overwhelmed by the variety of options and may not know where to start. This will also make practicing with the instrument more difficult, as there are many ways to use it, meaning it will require experimentation to figure out what works and what doesn't. This means that the barrier of entry is higher than some instruments, but more optimistically, that the possibility for virtuosity may also be higher, as users can master a wide variety of effect interactions and mapping possibilities if they can push past the initial confusion.

Another downside is that calibrating the Kinect will be slightly different for every user as their bodies and the upper and lower bounds of their 3D position values will be different. The Kinected DJ does work well for most individuals even when it is calibrated to a different individual, but for optimized use this effectively means that all users must recalibrate the instrument for themselves.

One final drawback is the effect patcher, which forces users to use Max (though briefly) if they want to change effects mid-performance. While it is still easy to change effects, this may cause the audience to lose immersion in a performance.

4.1 Next Steps

Mitigating the drawbacks of the Kinected DJ, as well as adding additional capabilities for performance, will be important in future iterations of this instrument. To address the issue of decision paralysis, it may be helpful to create presets for specific types of performance that users can start with and then branch out from once they feel comfortable with the technology. This would mean hand-selecting specific effect combinations and mappings for different performance needs.

For instance, if a dance DJ were to use the instrument, it may be desirable to limit the use of time-based effects and map more input to the hands and arms as dance music allows for more expressive use of these parts of the body. For an ambient or experimental performance, however, it may be more desirable to use experimental effects and to have mappings centered more on the body, as these performers may want more nuanced control over the effects they are using.

To solve the calibration issue, a more user-friendly system for calibration may be necessary with detailed instructions and easily understandable UI. It could also be desirable to include a generic model for calibration that would work reasonably well for most users while still giving users the option to calibrate for themselves if they wish to optimize their performance.

To solve the problem of the patcher, a matrix could be implemented which could then cycle through

presets using a MIDI device to minimize direct use of a computer. Matrixes can be confusing, however, so this may require a different solution to balance accessibility with a wide array of effect possibilities. Additional improvements could also be made to the general programming layout and UI to make for a cleaner and more accessible experience overall.

5. CONCLUSION

The Kinected DJ, while still in development, is an affordable and customizable digital music instrument that offers a great deal of performance possibilities through the full use of the body. While initially conceptualized for use by DJs, this device could be used in many musical environments, from singers, to instrumentalists, to re-imaging pre-existing pieces. With an almost infinite number of mapping possibilities, the Kinected DJ greatly expands the possibilities of expression, both for pre-existing instruments and more contemporary ways of manipulating sound.

Future work will focus on making the instrument more accessible to the public while retaining as much freedom to customize as possible. Conducting user studies will also be incredibly important for future work to address issues that may be missed in initial stages of development. With additional improvements, the Kinected DJ has the potential to be an accessible but customizable new instrument, and one that can allow for new methods of musical performance and expression for the 21st century.

6. REFERENCES

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